

UNIT II: NETWORK ARCHITECTURE AND IP ADDRESSING

Topics Covered

- **2.1** Network Standards, Ethernet Architecture, and Types of Ethernet
- **2.2** Client-Server vs. Peer-to-Peer Network Architecture
- **2.3** Internet, Intranet, and Extranet Concepts
- **2.4** Connection-Oriented vs. Connectionless Communication Services
- **2.5** IP Address Structure, Octets, and IPv4 Classes
- **2.6** Comprehensive Comparison: IPv4 vs. IPv6

2.1 NETWORK STANDARDS & ETHERNET ARCHITECTURE

Understanding Network Standards

Network standards define formal specifications, rules, procedures, and physical characteristics that ensure hardware and software manufactured by different vendors can interoperate seamlessly across different communication systems.

Types of Standards

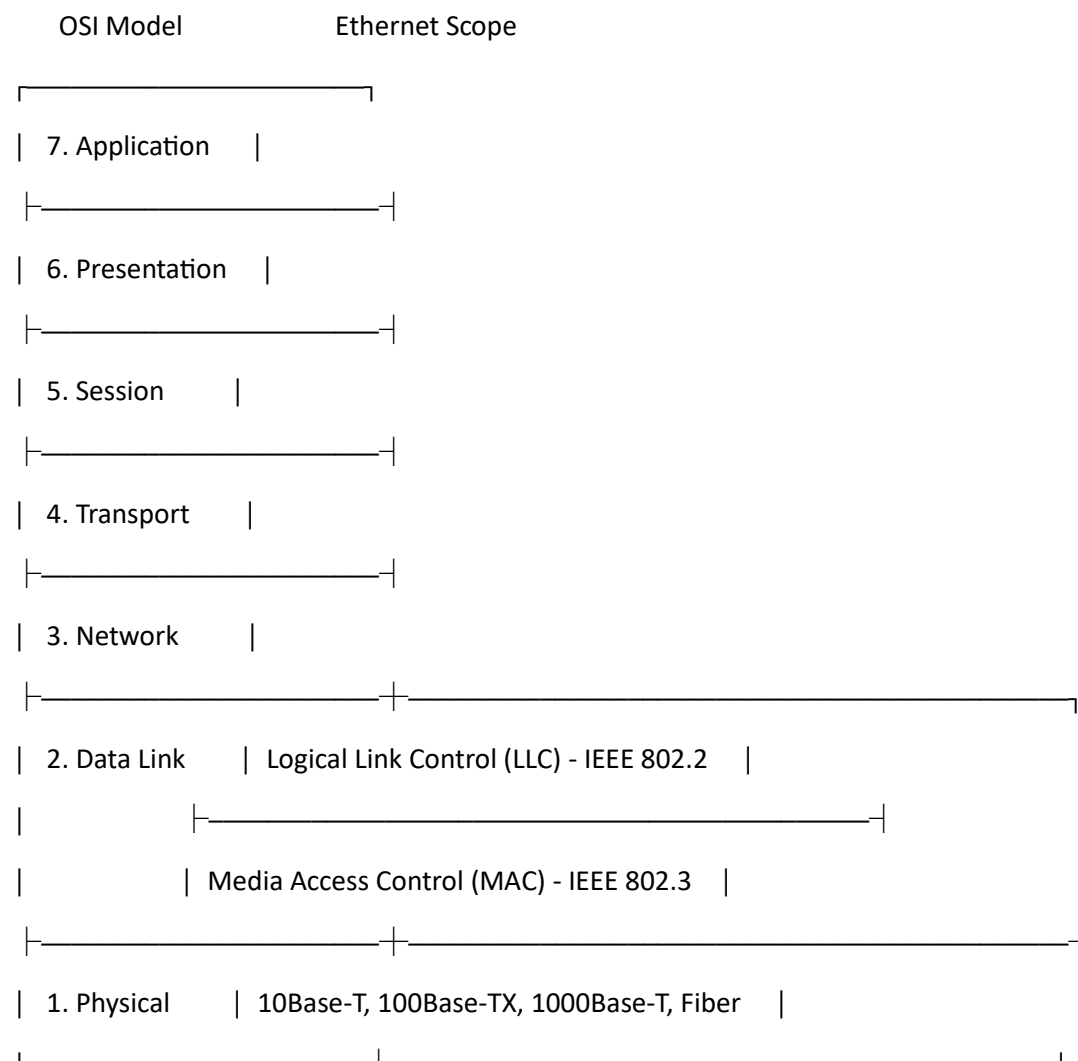
1. **De Facto Standards ("By Fact"):**
 - Standards that have emerged naturally through wide adoption, market popularity, or historical use without formal organization approval.
 - *Examples:* TCP/IP protocol suite, original IBM PC architecture.
2. **De Jure Standards ("By Law"):**
 - Official standards created, approved, and published by recognized standards bodies after formal review.
 - *Examples:* IEEE 802 specifications, OSI reference model.

Major Governing Standards Organizations

- **IEEE (Institute of Electrical and Electronics Engineers):** Defines LAN/MAN standards (e.g., IEEE 802.3 for Ethernet, IEEE 802.11 for Wi-Fi).
- **ISO (International Organization for Standardization):** Known for developing the 7-layer OSI Reference Model.
- **IETF (Internet Engineering Task Force):** Manages Internet protocols and publishes Requests for Comments (RFCs).
- **ITU-T (International Telecommunication Union - Telecommunication Standardization Sector):** Focuses on global telecommunication networks and video encoding standards.

What is Ethernet?

Ethernet is the most widely deployed wired Local Area Network (LAN) technology in the world. Defined primarily under the **IEEE 802.3** standard family, Ethernet operates at the **Physical Layer** and **Data Link Layer** (specifically the MAC sublayer) of the OSI model.



Key Characteristics of Ethernet

- **Frame-Based Structure:** Data is encapsulated into discrete frames containing source and destination MAC addresses, data payload, and error-checking fields (FCS).
- **CSMA/CD Mechanism:** Historically, shared Ethernet networks used **Carrier Sense Multiple Access with Collision Detection** to manage media access:
 1. *Carrier Sense:* Listen to the line before transmitting.
 2. *Multiple Access:* Devices share a common physical wire.
 3. *Collision Detection:* If two devices transmit simultaneously, a collision is detected, jamming signals are sent, and devices wait a random backoff time before retrying.
- **Modern Switch-Based Networks:** Contemporary Ethernet operates in **Full-Duplex** mode over dedicated switch connections, eliminating data collisions entirely and rendering CSMA/CD obsolete in modern setups.

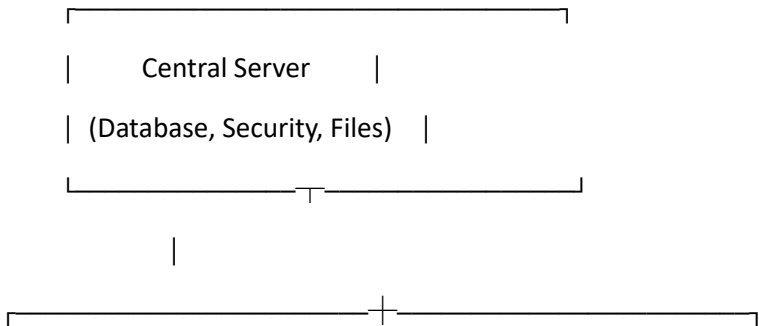
Comprehensive Types of Ethernet

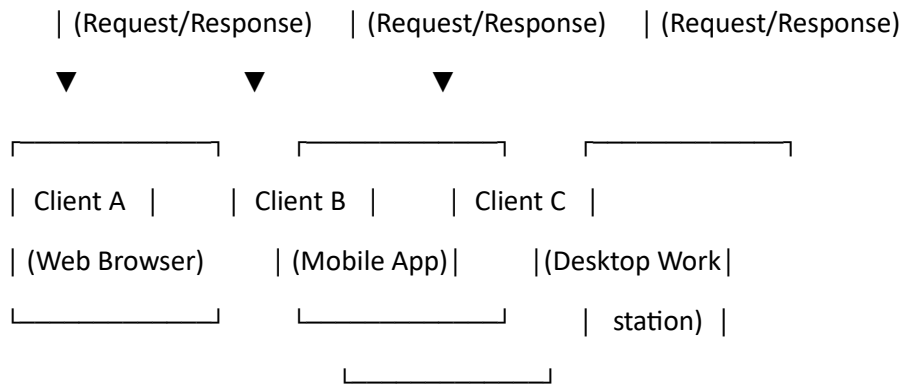
Ethernet Standard	IEEE Name	Speed	Medium / Cable Type	Max Segment Length	Primary Application
Standard Ethernet	802.3	10 Mbps	Coaxial (10Base2/10Base5) / UTP Cat3 (10Base-T)	100m – 500m	Legacy office LANs
Fast Ethernet	802.3u	100 Mbps	Twisted Pair Cat5 (100Base-TX) / Multi-mode Fiber (100Base-FX)	100m (UTP) / 2 km (Fiber)	Desktop connections
Gigabit Ethernet	802.3ab / 802.3z	1 Gbps (1000 Mbps)	UTP Cat5e/Cat6 (1000Base-T) / Fiber (1000Base-SX/LX)	100m (UTP) / 5 km (Fiber)	Modern LAN backbones & workstations
10-Gigabit Ethernet	802.3ae	10 Gbps	Cat6a/Cat7 UTP / Single-Mode Fiber (10GBase-LR)	100m (UTP) / 10 km – 40 km	Data centers & campus backbones
40/100-Gigabit Ethernet	802.3ba	40 / 100 Gbps	High-grade OM3/OM4 Optical Fiber / Twinax Cables	100m to 10 km+	Cloud providers & ISP backbones

2.2 CLIENT AND SERVER ARCHITECTURE

Overview

Network applications are typically organized around structured architectures that determine how tasks, storage, and processing powers are distributed across nodes.





Client-Server Architecture

In a **Client-Server Architecture**, tasks are split between central service providers (**Servers**) and service requesters (**Clients**).

- **Client:** A device or program (e.g., web browser, email software) that requests data, web pages, or application execution from a server.
- **Server:** A high-performance computer dedicated to listening for requests, executing business logic, processing queries, and serving resources back to clients.

Advantages

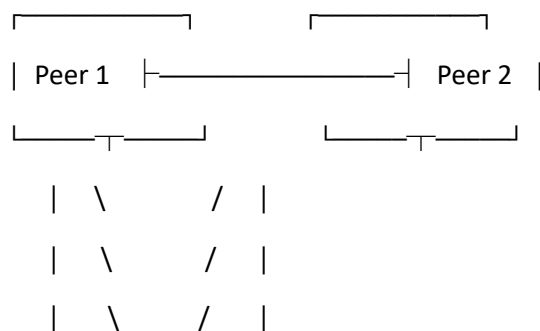
- **Centralized Data & Management:** Security, backups, and data access are maintained centrally.
- **High Security:** Access rights and data permissions are enforced at a central location.
- **Scalability:** Server capacity can be upgraded to handle growing client traffic.

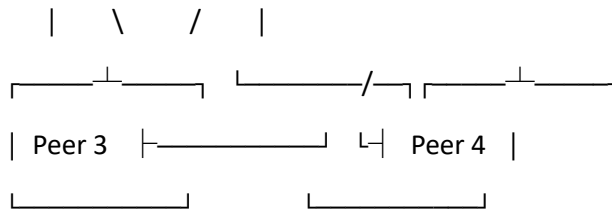
Disadvantages

- **Single Point of Failure:** If a central server goes offline, all connected clients lose access to central resources.
- **High Hardware Costs:** Specialized server hardware and software licensing are expensive.

Peer-to-Peer (P2P) Architecture

In a **Peer-to-Peer Architecture**, every node on the network (a "peer") acts as both a client and a server simultaneously. Nodes share their own computational power, bandwidth, and storage directly with other nodes without passing through a central server.





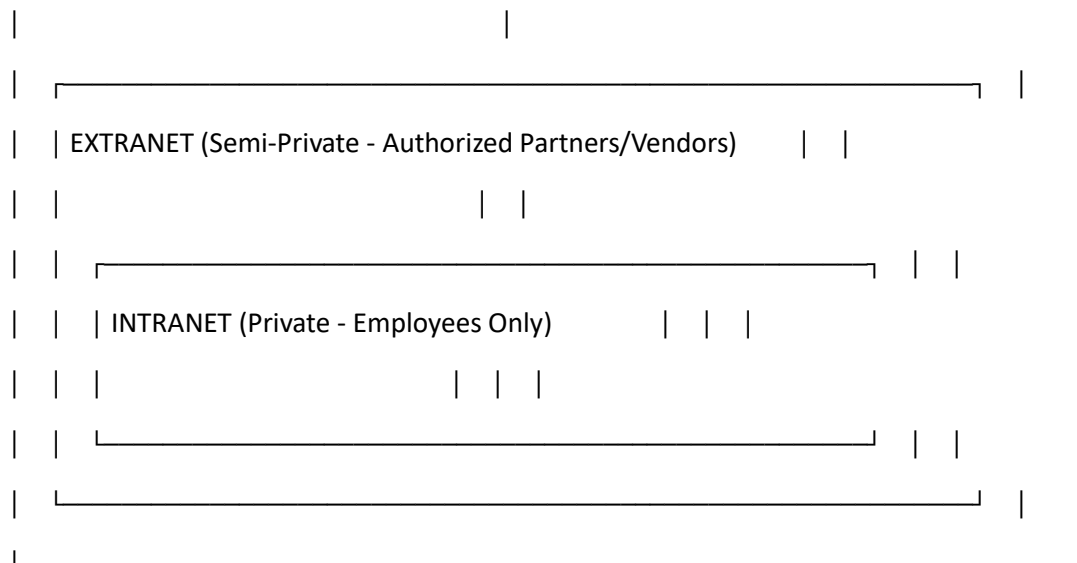
Detailed Comparison: Client-Server vs. Peer-to-Peer

Feature	Client-Server Architecture	Peer-to-Peer Architecture (P2P)
Centralization	Centralized architecture	Distributed / Decentralized architecture
Node Roles	Roles are strictly defined (Clients request, Servers respond)	Nodes perform dual roles (Client and Server simultaneously)
Data Storage	Stored centrally on dedicated server machines	Distributed across individual peer hard drives
Setup Cost	High (requires high-end servers & OS licenses)	Low (utilizes existing workstation hardware)
Scalability	Scaled vertically by upgrading server resources	Scaled horizontally (performance increases as nodes join)
Security & Administration	Highly secure; managed centrally by network admins	Harder to manage; relies on user-level security
Traffic Flow	Bottlenecks can occur at the central server	Traffic is distributed evenly across multiple peers
Examples	Web servers (HTTP), Email (SMTP/IMAP), Databases	BitTorrent, Skype (legacy), Blockchain networks

2.3 INTERNET, INTRANET, AND EXTRANET

Network environments are categorized based on their accessibility, security boundaries, and audience reach.

INTERNET (Public - Worldwide Access)	
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1. Internet

The **Internet** is a global system of interconnected computer networks operating on the standardized **TCP/IP** protocol suite. It links billions of devices worldwide across public, academic, corporate, and government networks.

- **Access:** Fully public; open to anyone with an Internet Service Provider (ISP).
- **Security:** Low baseline security; users rely on end-to-end encryption (e.g., HTTPS, TLS).

2. Intranet

An **Intranet** is a private, isolated network deployed within an organization to allow staff to securely communicate, collaborate, and store company documents.

- **Access:** Strictly private; accessible only by authorized employees inside the enterprise network or via a secure Virtual Private Network (VPN).
- **Security:** High security; protected from public view behind corporate firewalls, intrusion prevention systems, and authentication gateways.

3. Extranet

An **Extranet** is a controlled extension of a private Intranet that grants selective access to trusted external entities, such as business partners, suppliers, vendors, or specific customers.

- **Access:** Semi-private; access is restricted to verified external users.
- **Security:** Very high; uses encrypted tunnels (VPNs), multi-factor authentication (MFA), and granular access rules.

Detailed Comparison Table

Property	Internet	Intranet	Extranet
Target Audience	General public worldwide	Internal company employees	Employees, vendors, partners, and clients
Network Type	Public	Private	Semi-Private
Accessibility	Unrestricted worldwide access	Strictly inside the organization	Limited external access via portal or VPN
Security Level	Public / Unsecured by default	Highly secured behind firewalls	Securely isolated via VPNs & credentials
Primary Purpose	Sharing general information globally	Internal enterprise operations & workflow	Inter-organizational business collaboration
Example	Browsing Wikipedia or Google	Company HR portal, internal payroll	Supplier supply-chain management portal

2.4 CONNECTION-ORIENTED VS. CONNECTIONLESS SERVICES

Data transfer at the Transport and Network layers occurs via two fundamental operational paradigms: **Connection-Oriented** or **Connectionless** services.

A. Connection-Oriented Service

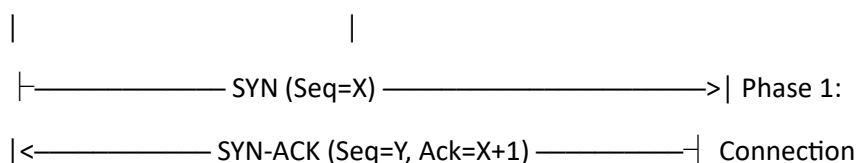
Connection-oriented communication establishes a logical virtual circuit between the sender and receiver before transmitting any payload data.

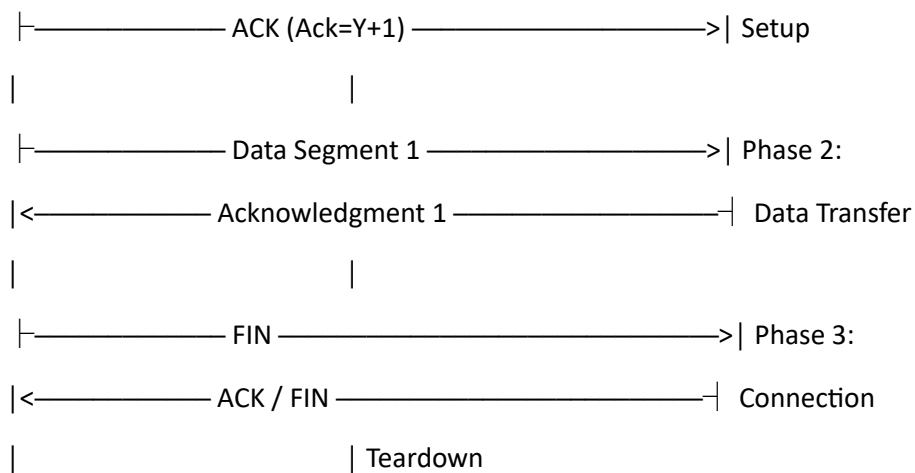
The Three-Phase Lifecycle

1. **Connection Establishment:** Devices perform a handshake to negotiate session parameters (e.g., sequence numbers, buffer sizes).
2. **Data Transfer:** Payload data is split into segments, transmitted sequentially, acknowledged, and reassembled.
3. **Connection Termination:** Devices gracefully teardown the channel to free system buffers.

Sender (Client)

Receiver (Server)





- **Key Features:** Reliable, sequential delivery, flow control (sliding window), error recovery (ARQ), and congestion handling.
- **Primary Example:** TCP (Transmission Control Protocol).

B. Connectionless Service

Connectionless communication sends data packets independently without prior handshaking or checking if the recipient is active and reachable.

- **Datagram Concept:** Each packet (datagram) contains complete source and destination addressing information and is routed independently through the network.
- **Key Features:** Unreliable ("best-effort"), zero setup delay, low header overhead, no sequence numbering or acknowledgments.
- **Primary Example:** UDP (User Datagram Protocol), IP (Internet Protocol).

In-Depth Comparison Table

Feature Parameter	Connection-Oriented Service	Connectionless Service
Setup Phase	Explicit setup (Handshake) required	No initial setup phase
Primary Protocol	TCP (Transport Control Protocol)	UDP (User Datagram Protocol)
Reliability	Guaranteed: Lost packets are retransmitted	Best-effort: Lost packets are dropped
Data Ordering	In-Sequence: Packets arrive in order	Out-of-Order: Packets can arrive out of sequence

Feature Parameter	Connection-Oriented Service	Connectionless Service
Header Size Overhead	Large header (minimum 20 bytes for TCP)	Small header (fixed 8 bytes for UDP)
Flow & Congestion Control	Yes (sliding window and congestion algorithms)	None
Speed & Latency	Slower due to setup phase and ACK delays	Extremely fast with minimal transmission delay
Communication Type	Point-to-point stream	Unicast, Multicast, and Broadcast
Ideal Applications	File Transfer (FTP), Web (HTTP/HTTPS), Email (SMTP)	Live Video Streaming, Online Gaming, DNS, VoIP

2.5 IP ADDRESS CLASSES (IPv4)

An **Internet Protocol Version 4 (IPv4)** address is a 32-bit binary number assigned to identify network interfaces on a TCP/IP network.

Notation and Structure

- **Format:** Written in **dotted-decimal notation** comprising four decimal values (each ranging from 0 to 255), separated by periods.
- **Octet Division:** An IPv4 address contains 4 octets (8 bits per octet):

$$32 \text{ Bits Total} = \underbrace{8 \text{ bits}}_{\text{Octet 1}} . \underbrace{8 \text{ bits}}_{\text{Octet 2}} . \underbrace{8 \text{ bits}}_{\text{Octet 3}} . \underbrace{8 \text{ bits}}_{\text{Octet 4}}$$

- **Logical Components:**
 1. **Network ID (Net ID):** Identifies the overall network segment assigned by authority bodies.
 2. **Host ID (Host ID):** Identifies the unique device interface on that network.

IPv4 Address Classes

To manage global address allocation, IPv4 addresses were originally divided into 5 distinct classes based on the leading bits of the first octet:

1st Byte (First Octet) Structure

Class A: 0 ——— Network (7 bits) ——— | ——— Host ID (24 bits) ———

Class B: 10 ——— Network (14 bits) ——— | ——— Host ID (16 bits) ———

Class C: 110 ——— Network (21 bits) ——— | ——— Host ID (8 bits) ———

Class D: 1110—— Multicast Group ID (28 bits) _____

Class E: 1111—— Experimental / Reserved (28 bits) _____

Detailed Class Breakdown

Class	High-Order Bits	First Octet Range	Default Subnet Mask	Net ID / Host ID Split	Total Available Networks	Usable Hosts per Network	Primary Application
Class A	0	1 – 127 *	255.0.0.0 (/8)	8 bits Net ID / 24 bits Host ID	126	$2^{24} - 2 = 16,777,214$	Extremely large corporations, ISPs, & governments
Class B	10	128 – 191	255.255.0.0 (/16)	16 bits Net ID / 16 bits Host ID	16,384	$2^{16} - 2 = 65,534$	Medium-sized enterprises, universities
Class C	110	192 – 223	255.255.255.0 (/24)	24 bits Net ID / 8 bits Host ID	2,097,152	$2^8 - 2 = 254$	Small businesses, home LANs
Class D	1110	224 – 239	Undefined	Multicast addressing	N/A	N/A	Video broadcasting, routing protocol updates
Class E	1111	240 – 255	Undefined	Reserved	N/A	N/A	R&D, experimental testing

Notes on Available Host Calculation:

- For any network, two host addresses are reserved and unusable for endpoints:

1. **Network Address:** Where all Host ID bits are 0s (e.g., 192.168.1.0).
 2. **Broadcast Address:** Where all Host ID bits are 1s (e.g., 192.168.1.255).
- **Formula for Usable Hosts:** $2^H - 2$ (where H represents the number of host bits).

Special IPv4 Addresses

1. **Loopback Address Range (127.0.0.0 to 127.255.255.255):**
 - Reserved for local testing and diagnostic functions.
 - 127.0.0.1 (**localhost**) routes traffic directly back to the local device's internal software stack without hitting the physical network interface card (NIC).
2. **Private IP Address Ranges (RFC 1918):**
 - These addresses are non-routable over the public Internet and are reserved for internal local area networks.

Class	Private Address Range	Total CIDR Block	Total Addresses
Class A	10.0.0.0 – 10.255.255.255	10.0.0.0/8	16,777,216
Class B	172.16.0.0 – 172.31.255.255	172.16.0.0/12	1,048,576
Class C	192.168.0.0 – 192.168.255.255	192.168.0.0/16	65,536

2.6 IPV4 VS. IPV6

The Transition Driver: Exhaustion of IPv4

IPv4 uses a 32-bit address space, allowing for $2^{32} \approx 4.29$ billion distinct addresses. With the explosive rise of mobile devices, IoT equipment, smart appliances, and global internet expansion, available unassigned IPv4 addresses were exhausted.

To address this limitation, the IETF created **IPv6**, which employs a **128-bit** address space.

Address Format and Representation

- **IPv4 Representation:** Written in decimal format, split into 4 octets separated by dots.
 - *Example:* 192.168.1.1
- **IPv6 Representation:** Written in hexadecimal format, split into 8 blocks (hextets) of 4 hex digits separated by colons.
 - *Example:* 2001:0db8:85a3:0000:0000:8a2e:0370:7334

IPv6 Address Shortening Rules

1. **Omit Leading Zeros:** Leading zeros within a hextet can be dropped.
 - 0042 becomes 42

2. **Zero Compression (::):** Consecutive blocks of zeros can be replaced with a single double colon (::). *This rule can only be applied once per address.*

- 2001:0db8:85a3:0000:0000:8a2e:0370:7334 simplifies to
2001:db8:85a3::8a2e:370:7334

Exhaustive Architectural Comparison: IPv4 vs. IPv6

Feature Parameter	IPv4	IPv6
Address Length	32 bits (4 bytes)	128 bits (16 bytes)
Address Space Size	$2^{32} \approx 4.29 \times 10^9$ addresses	$2^{128} \approx 3.4 \times 10^{38}$ addresses
Address Notation	Dotted-Decimal (e.g., 172.16.254.1)	Hexadecimal with Colons (e.g., fe80::1)
Header Format	Variable length (20 to 60 bytes)	Fixed length (40 bytes)
Checksum Field	Included in header; recalculated at every router hop	Removed from header; relies on Layer 2 and Layer 4 checksums
Address Assignment	Manual or dynamic via DHCP	Auto-configuration via SLAAC (Stateless) or DHCPv6
Packet Fragmentation	Handled by intermediate routers and senders	Handled only by the sending device
Security (IPsec)	Optional extra feature	Built-in native architectural component
Transmission Modes	Unicast, Multicast, and Broadcast	Unicast, Multicast, and Anycast (No Broadcast)
Quality of Service	Uses ToS (Type of Service) / DiffServ fields	Uses dedicated Flow Label and Traffic Class fields

QUICK REVISION & SUMMARY

Key Technical Terms

- **Ethernet (IEEE 802.3):** The dominant frame-based physical and data-link layer standard for wired LANs.
- **CSMA/CD:** Media access control method historically used by shared Ethernet to detect and handle frame collisions.
- **Client-Server Architecture:** Centralized system model where clients request processing services from dedicated servers.
- **Intranet:** An isolated, secure internal company network accessible only to employees.
- **Extranet:** An extension of an internal Intranet providing controlled access to trusted partners and vendors.
- **TCP (Transmission Control Protocol):** Connection-oriented, reliable transport layer protocol with flow control.
- **UDP (User Datagram Protocol):** Connectionless, lightweight transport layer protocol designed for real-time traffic.
- **IPv4 Address:** A 32-bit dotted-decimal logical identifier divided into network and host portions across classes A–E.
- **IPv6 Address:** A 128-bit colon-hexadecimal address space developed to eliminate global IP address exhaustion.

IMPORTANT EXAM QUESTIONS

Short Answer Questions

1. Differentiate between *De Facto* and *De Jure* network standards with real-world examples.
2. What is CSMA/CD, and why is it not required in modern full-duplex Ethernet networks?
3. Calculate the usable host capacity for a Class B IPv4 network using the standard formula.
4. Define the loopback IP address and state its exact address range.
5. Why was the header checksum field removed in the IPv6 header structure?
6. Explain the zero-compression rule (::) used in IPv6 address notation.

Long Answer Questions

1. **Ethernet Analysis:** Discuss the evolution of Ethernet standards from Standard Ethernet (10 Mbps) to 10-Gigabit Ethernet. Include details on physical media, speeds, segment lengths, and frame processing.
2. **Architecture Comparison:** Compare Client-Server and Peer-to-Peer network architectures across scalability, hardware cost, administration, security, and failure resistance.
3. **Services Analysis:** Describe Connection-Oriented and Connectionless services. Include a diagram showing the TCP 3-way handshake and contrast it with UDP datagram delivery.
4. **IP Addressing Deep-Dive:** Outline the 5 IPv4 Address Classes. Provide first-octet binary ranges, default subnet masks, and formulas for calculating net IDs and host capacities.

5. **Protocol Migration:** Provide a full comparison between IPv4 and IPv6 across address formats, total address space, header layout, security implementations, and transmission mechanisms.